



WOLDIA UNIVERSITY

COLLEGE OF HEALTH SCIENCES

**ASSESSMENT OF KNOWLEDGE, ATTITUDE, AND PRACTICE
TOWARDS VITAMIN D AMONG MOTHERS OF UNDER-FIVE
CHILDREN IN SEKOTA TOWN AND SEKOTA ZURIA, AMHARA
REGION, ETHIOPIA, 2026. A COMMUNITY BASED CROSS-
SECTIONAL COMPARATIVE STUDY.**

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**A THESIS PROPOSAL SUBMITTED TO THE DEPARTMENT OF
PUBLIC HEALTH COLLEGE OF HEALTH SCIENCES, WOLDIA
UNIVERSITY IN PARTIAL FULFILMENT OF THE DEGREE OF
MASTER OF PUBLIC HEALTH IN HUMAN NUTRITION.**

JULY. 2026

WOLDIA UNIVERSITY, ETHIOPIA

APPROVAL SHEET
COLLEGE OF HEALTH SCIENCES
DEPARTMENT OF PUBLIC HEALTH

I, the undersigned MPH student, declare that this research proposal is my original work in partial fulfillment of the requirements for the Master of Public Health.

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I certify that I have read and evaluated this research proposal entitled with: Assessment of Knowledge, Attitude, and Practice towards Vitamin D among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia,2026. A Community Based Cross-Sectional Comparative Study.

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ACKNOWLEDGMENT

First, I would like to thank the Department of Public Health, College of Health Sciences, Woldia University for providing me with this opportunity.

I would also like to thank God for everything. I am also deeply indebted to my advisors, Dr. Melese Abate Reta (PhD), for their unreserved support and constructive comments during the preparation of this proposal. Without their advice, guidance, and patience, the completion of this proposal would have been impossible.

I would also like to extend my sincere appreciation to the Sekota Town and Sekota Zuria Woreda Health Offices, the study participants, data collectors, and supervisors for their cooperation and support during the data collection process. Their willingness to participate and contribute to this research is highly valued.

Finally, I would like to express my sincere gratitude to the head of my organization and my co-workers for their encouragement and support in enabling me to pursue my MPH studies.

ABBREVIATIONS AND ACRONYMS

ANC	Antenatal Care
AOR	Adjusted Odds Ratio
CI	Confidence Interval
COR	Crude Odds Ratio
CSA	Central Statistical Agency
DHS	Demographic and Health Survey
EDHS	Ethiopian Demographic and Health Survey
EPHI	Ethiopian Public Health Institute
ETB	Ethiopian Birr
FMoH	Federal Ministry of Health
HEWs	Health Extension Workers
HSTP	Health Sector Transformation Plan
KAP	Knowledge, Attitude and Practice
NNP	National Nutritional Program
PNC	Postnatal Care
SDG	Sustainable Development Goals
SPSS	Statistical Package for Social Sciences
UVB	Ultraviolet B
VDD	Vitamin D Deficiency
WHO	World Health Organization

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ABSTRACT

Background: Vitamin D deficiency is a growing public health concern globally, affecting approximately one billion people worldwide (30%–50% of the global population). In Ethiopia, national-level data are limited; however, facility-based and regional studies indicate that vitamin D deficiency among children may range between 30% and 45% in some settings. Maternal knowledge, attitude, and practice (KAP) significantly influence children's vitamin D status. However, there is limited evidence comparing rural and urban maternal KAP in Ethiopia, particularly in Sekota town.

Objective: To assess and compare knowledge, attitude, and practice towards vitamin D among mothers of under-five children in rural and urban areas of Sekota town and Sekota Zuria, Amhara Region, Ethiopia, 2026.

Methods: A community-based comparative cross-sectional study was conducted among 548 mothers of under-five children in rural and urban areas of Sekota, Amhara Region, Ethiopia. A multistage stratified sampling technique was used. Data were collected through a structured interviewer-administered questionnaire, entered into EpiData version 4.6, and exported to SPSS version 26 for analysis. Descriptive statistics, binary and multivariable logistic regression, and Chi-square tests were used for analysis. Mean scores were calculated, and respondents scoring $\geq$ mean were categorized as having good knowledge, positive attitude, and good practice.

Results: The study found that 71.0% of mothers had good knowledge, 58.6% had favorable attitudes, and 48.4% demonstrated good practice regarding vitamin D. Urban mothers had significantly higher good knowledge (86.5% vs 55.5%, $p < 0.001$) and good practice (54.0% vs 42.7%, $p = 0.008$) compared to rural mothers. Residence, maternal education, ANC attendance, media exposure, and attitude were significant predictors of knowledge. Knowledge, practice, and PNC attendance were predictors of attitude, while residence and attitude were predictors of practice.

Conclusion: While the majority of mothers had good knowledge and favorable attitudes, practice remained suboptimal. Significant urban-rural disparities exist, with urban mothers having better outcomes. Targeted health education interventions focusing on safe sun exposure practices and addressing misconceptions are needed to bridge the knowledge-practice gap and improve vitamin D status among children in Sekota.

Keywords: *Vitamin D, Knowledge, attitude, and practice, under-five children, rural-urban comparison, Sekota*

1. INTRODUCTION

1.1 Background

Vitamin D is a fat-soluble steroid hormone that plays a critical role in calcium and phosphorus homeostasis, skeletal development, immune regulation, and cellular growth (1; 2; 3) . It is primarily synthesized in the skin through exposure to ultraviolet B (UVB) radiation from sunlight and can also be obtained from limited dietary sources such as fatty fish, egg yolk, fortified foods, and supplements (3; 4).

In children, vitamin D is essential for proper bone mineralization and growth (5). Deficiency during early childhood can lead to rickets, delayed motor development, bone deformities, impaired growth, and increased susceptibility to infections (5). Emerging evidence also links vitamin D deficiency with respiratory infections, autoimmune diseases, and impaired immune function (1; 5).

Globally, vitamin D deficiency has become a significant public health concern affecting approximately one billion people, about 30% to 50% global population (2; 3). In Ethiopia national level data are limited; however, facility based and regional studies indicate that vitamin D deficiency among children may range between 30% and 45% in some settings (6). Surprisingly, the problem is not limited to cold or less sunny countries, even in tropical and subtropical regions, including Africa; vitamin D deficiency is widely reported (7). Factors contributing to this paradox include limited sun exposure due to indoor lifestyles, cultural clothing practices, air pollution, poor dietary diversity, and lack of awareness about vitamin D (3; 7).

Under-five children are particularly vulnerable because of rapid growth demands and dependence on caregivers for feeding and outdoor exposure. Mothers, as primary caregivers, play a central role in determining children's dietary intake, sun exposure practices, healthcare-seeking behavior, and supplementation practices. Therefore, maternal knowledge, attitude, and practice (KAP) regarding vitamin D are critical determinants of children's vitamin D status (8; 9).

In Ethiopia, national nutrition programs strongly emphasize vitamin A, iron, and iodine deficiencies. However, vitamin D receives limited policy and program attention despite increasing evidence of micronutrient deficiencies among children (10; 11). In areas like Sekota Town, and Sekota Zuria Woreda, Wag-Himra Zone where food insecurity, child under nutrition, and recurrent drought are common micronutrient deficiencies may be exacerbated.

Despite these concerns, there is limited empirical evidence assessing maternal awareness and practices related to vitamin D, particularly comparing rural and urban populations. Understanding rural–urban differences is important because educational access, health service utilization, exposure to health information, and lifestyle patterns vary significantly between these settings.

Therefore, this study aims to assess and compare the knowledge, attitude, and practice toward vitamin D among mothers of under-five children in rural and urban areas of Sekota, Amhara Region, Ethiopia.

1.2 Statement of the Problem

Micronutrient deficiencies often referred to as "hidden hunger," affect more than two billion people globally and contribute substantially to childhood morbidity, mortality, and long-term developmental impairment (12). Among these, vitamin D deficiency has emerged as a significant but often overlooked public health concern, affecting approximately 30–50% of the global population across all age groups (1; 2).

Unlike vitamin A, iron, and iodine deficiencies, which have long been prioritized in global and national nutrition programs, vitamin D deficiency remains under-recognized despite its serious health consequences, particularly for children under five years of age (13).

Vitamin D plays an essential role in calcium homeostasis, bone mineralization, immune function, and cellular growth. In under-five children, deficiency can lead to rickets, delayed motor development, bone deformities, growth stunting, and increased susceptibility to acute respiratory infections (5). Emerging evidence also links vitamin D deficiency with impaired immune function, increased risk of autoimmune diseases, and higher all-cause childhood mortality (1).

The World Health Organization estimates that micronutrient deficiencies, including vitamin D deficiency, account for approximately 7–10% of the total disease burden in low- and middle-income countries (12). In Ethiopia, where under-five mortality remains a national priority, addressing preventable causes of childhood morbidity such as vitamin D deficiency is critical for achieving the Sustainable Development Goals (SDG 3.2) (14) and the Ethiopian Health Sector Transformation Plan (HSTP II) (15).

Ethiopia is located in the tropics and receives abundant sunlight throughout the year, which theoretically should provide sufficient ultraviolet B (UVB) radiation for endogenous vitamin

D synthesis. However, studies from Ethiopia and other African countries have documented significant vitamin D deficiency among children and pregnant women, with prevalence ranging from 30% to 45% in some settings (6; 16).

This paradox is explained by multiple interacting factors: limited sun exposure due to indoor lifestyles and cultural clothing practices, poor dietary diversity, low consumption of vitamin D-rich foods such as fish and eggs, absence of food fortification programs, and lack of routine supplementation (7; 10). These factors are not merely biomedical but are deeply embedded in social, cultural, economic, and health system contexts that vary significantly between rural and urban populations.

Mothers, as primary caregivers, play a pivotal role in determining children's vitamin D status through their decisions regarding sun exposure, dietary practices, and healthcare-seeking behaviour. Maternal knowledge about vitamin D sources, benefits, and deficiency consequences directly influences whether children receive adequate sunlight exposure, consume vitamin D-rich foods, or receive supplements when indicated (8; 17). Furthermore, maternal attitudes/beliefs about sun exposure, perceptions of supplementation, and cultural norms shape daily practices that either promote or undermine children's vitamin D adequacy.

In Ethiopia, studies have shown that while many mothers are aware of the general benefits of sun exposure, significant gaps exist in knowledge about appropriate timing, duration, and frequency of exposure, as well as dietary sources of vitamin D (8). Moreover, misconceptions persist, such as the belief that breast milk alone provides sufficient vitamin D or that covering infants completely during sun exposure is beneficial. These knowledge gaps and misconceptions represent modifiable determinants that can be addressed through targeted health education.

Ethiopia is characterized by profound rural-urban disparities in health outcomes, health service utilization, educational attainment, media exposure, and lifestyle patterns. According to the Ethiopian Mini Demographic and Health Survey (EDHS), urban women have significantly higher rates of antenatal care attendance (84% vs. 43%), skilled birth attendance (78% vs. 28%), and media exposure (68% vs. 32%) compared to rural women (9). Conversely, rural children may have greater natural sun exposure due to outdoor livelihoods but face higher rates of food insecurity, dietary inadequacy, and limited access to health information.

These differences have important implications for vitamin D-related knowledge, attitudes, and practices. Urban mothers may have better access to health information but may keep children

indoors more frequently due to housing conditions, safety concerns, or lifestyle factors. Rural mothers may have less formal knowledge but may practice longer sun exposure. Understanding these contextual differences is essential for designing effective, context-specific interventions. Yet, very few studies in Ethiopia have systematically compared rural and urban maternal KAP regarding vitamin D.

Ethiopia has made significant progress in addressing micronutrient deficiencies through national programs such as the National Nutrition Program (NNP II), vitamin A supplementation campaigns, iron-folate supplementation for pregnant women, and salt iodization (18). However, vitamin D has received minimal policy attention. It is not included in routine maternal and child health education, nor is it part of essential nutrition packages delivered by Health Extension Workers (HEWs) (18). The Ethiopian Food and Nutrition Policy (2018) and the National Nutrition Program II (2016–2020) focus primarily on protein-energy malnutrition, vitamin A, iron, and iodine, with no explicit mention of vitamin D. This policy neglect means that mothers receive little to no guidance on vitamin D during antenatal care, postnatal care, or child health services (19; 18).

Sekota, located in Wag Himra Zone of the Amhara Region, is one of Ethiopia's most food-insecure areas, with recurrent drought, chronic malnutrition, and limited health infrastructure. Evidence from Wag-Himra zone indicates that under-five malnutrition remains a serious public health problem. In Sekota area stunting affects more than half of children (about 51-55%) while acute malnutrition ranges around 8-12% (20). In such settings, micronutrient deficiencies are likely to be exacerbated by food insecurity, limited dietary diversity, and reduced access to health services. However, no study has been conducted in Sekota to assess maternal knowledge, attitudes, and practices regarding vitamin D. Without local evidence, health authorities cannot design targeted interventions, and HEWs lack guidance on integrating vitamin D education into routine services.

While global and regional studies have documented poor maternal knowledge and suboptimal practices regarding vitamin D, there is a critical gap in community-based evidence from Ethiopia, particularly in rural areas and comparative studies between rural and urban settings. In Wag Himra Zone and Sekota specifically, no published study has assessed maternal KAP toward vitamin D among mothers of under-five children. Furthermore, the absence of rural-urban comparative data limits the ability of health planners to design context-appropriate

interventions. Without such evidence, vitamin D deficiency were likely remain a "hidden" contributor to childhood morbidity and mortality in this vulnerable population.

In summary, vitamin D deficiency among under-five children represents a significant but under-recognized public health problem in Ethiopia, with serious consequences for child health and development. Maternal knowledge, attitudes, and practices are key modifiable determinants, yet remain poorly understood, particularly in rural areas and in comparison with urban settings. In Sekota, where food insecurity, malnutrition, and limited health services converge, the lack of local evidence hinders effective program planning. This study aims to fill this critical gap by assessing and comparing maternal KAP toward vitamin D in rural and urban Sekota, thereby providing actionable evidence for nutrition education interventions and contributing to broader efforts to improve child health in Ethiopia.

1.3 Significance of the Study

This study were provide important evidence regarding maternal knowledge, attitude, and practice (KAP) toward vitamin D among mothers of under-five children in Sekota town and surrounding rural areas. Mothers play a central role in determining children's sun exposure, dietary intake, and health-seeking behaviour's, which directly influence children's vitamin D status and overall health. Therefore, understanding the level of maternal awareness and practices related to vitamin D is essential for designing effective interventions.

The findings were help identify specific gaps in maternal knowledge and inappropriate practices related to vitamin D, including sunlight exposure and dietary sources. This information was directly useful for the Sekota Woreda Health Office and local health facilities to design targeted nutrition education and health promotion programs for mothers and caregivers. In particular, the comparative analysis between rural and urban mothers were provide valuable insights into how residence, access to health services, and exposure to health information influence KAP. This evidence enables policymakers and health professionals to develop context-specific interventions that address the unique needs of both rural and urban communities.

At the national level, the study were contribute to the limited body of scientific evidence on vitamin D awareness and practices in Ethiopia, where vitamin D remains largely absent from routine maternal and child health guidelines. The findings may assist governmental and non-governmental organizations working in nutrition and child health to develop strategies for preventing vitamin D deficiency among children. Finally, this study serves as a baseline and

reference for future research in maternal and child nutrition, supporting advocacy for integrating vitamin D education into Ethiopia's National Nutrition Program and Essential Nutrition Actions framework.

2. LITERATURE REVIEW

2.1 Global overview of vitamin D deficiency

Vitamin D is an essential fat-soluble vitamin that plays a crucial role in calcium and phosphorus metabolism, bone mineralization, immune regulation, and cellular growth (1; 3). It is primarily synthesized in the skin following exposure to ultraviolet-B (UVB) radiation from sunlight, while a smaller proportion is obtained from dietary sources such as fatty fish, egg yolk, fortified foods, and supplements (3; 4).

Vitamin D deficiency has become a global public health problem affecting populations across different geographic regions. It is estimated that approximately one billion people worldwide suffer from vitamin D deficiency or insufficiency (3). According to global estimates, nearly 30–50% of the world's population may have insufficient vitamin D levels, making it one of the most common micronutrient deficiencies globally (2).

Historically, vitamin D deficiency was first identified in the early twentieth century when rickets became widespread among children living in industrialized cities in Europe and North America due to limited sun exposure and poor dietary intake (15; 16). Public health interventions such as cod liver oil supplementation and food fortification significantly reduced the prevalence of rickets by the mid-twentieth century (13). However, in recent decades, vitamin D deficiency has re-emerged in many countries due to urbanization, indoor lifestyles, and reduced outdoor activities (17).

Several systematic reviews have confirmed the widespread prevalence of vitamin D deficiency worldwide. A systematic review examining vitamin D status across multiple countries found that a large proportion of populations had serum vitamin D levels below recommended levels (18). Similarly, another review highlighted that vitamin D deficiency is prevalent not only in temperate regions but also in tropical and subtropical countries with abundant sunlight (2).

These findings indicate that environmental availability of sunlight alone does not guarantee adequate vitamin D status, as behavioural, cultural, and dietary factors strongly influence vitamin D synthesis and intake.

2.2 Maternal Knowledge, Attitude, and Practice toward Vitamin D

Maternal knowledge, attitude, and practice (KAP) regarding vitamin D are critical determinants of children's vitamin D status. Mothers play a central role in determining children's dietary

intake, sun exposure practices, and health-seeking behaviour's. Therefore, inadequate maternal awareness can contribute to increased risk of vitamin D deficiency among children.

Several studies conducted in different countries have reported insufficient maternal knowledge regarding vitamin D. A community-based study conducted in India among mothers of young children reported that approximately two-thirds of mothers had poor knowledge about vitamin D, and only a minority were aware that sunlight is the primary natural source of vitamin D (19). Similarly, another study conducted in India reported that more than 60% of mothers lacked adequate knowledge regarding the importance of vitamin D and appropriate sun exposure practices (20).

Studies conducted in the Middle East also highlight similar findings. A hospital-based cross-sectional study conducted in Saudi Arabia found that although many mothers had heard about vitamin D, a large proportion avoided sun exposure due to cultural practices and fear of skin damage (21). Another study conducted among Saudi mothers reported that while nearly 70% of mothers were aware of vitamin D, only about one-third practiced appropriate sun exposure for their children (22).

Evidence from Southeast Asia indicates that maternal education plays an important role in determining knowledge and practice related to vitamin D. A cross-sectional study conducted in Malaysia among mothers of young children found that higher maternal educational status was significantly associated with better knowledge of vitamin D sources and benefits. However, the study also reported that urban mothers tended to expose their children to sunlight less frequently due to indoor lifestyle patterns (23).

These findings demonstrate that although awareness of vitamin D may exist in some populations, appropriate practices related to sun exposure and dietary intake are often inadequate.

2.3 Evidence from Recent Global Studies

Recent global studies have further emphasized the persistence of vitamin D deficiency and the role of behavioural and lifestyle factors in its occurrence. A recent systematic review analysing global vitamin D status among children reported that vitamin D deficiency remains highly prevalent in many low- and middle-income countries, particularly in urban populations (24). Reduced outdoor activity, sedentary lifestyles, and limited dietary diversity were identified as major contributing factors.

Similarly, a cross-sectional study conducted in Turkey among mothers of under-five children found that although a large proportion of mothers had heard about vitamin D, only about 30%

practiced appropriate sun exposure timing for their children. The study also reported that maternal education level and counselling during antenatal care visits were significantly associated with improved knowledge and practice related to vitamin D (25).

These findings highlight the importance of maternal education and health service utilization in improving vitamin D awareness and related practices.

2.4 Evidence from Africa

For many years, it was assumed that vitamin D deficiency was uncommon in Africa because of abundant sunlight exposure throughout the year. However, recent studies have challenged this assumption and demonstrated that vitamin D deficiency can occur even in sunny regions due to lifestyle, cultural, and dietary factors.

A cross-sectional study conducted among pregnant women in Nigeria found that a significant proportion of mothers had inadequate knowledge regarding the importance of vitamin D and appropriate sun exposure practices. The study also reported poor dietary and behavioural practices related to vitamin D among participants (26).

Similarly, a comparative study conducted in Kenya among rural and urban mothers found notable differences in maternal knowledge and practices regarding vitamin D. The study reported that urban mothers generally had better knowledge but poorer sun exposure practices compared to rural mothers, mainly due to urban lifestyle patterns and reduced outdoor activity (10).

Another study conducted in South Africa assessing vitamin D status among children found that children living in urban areas had significantly lower vitamin D levels compared to those living in rural settings. The study also indicated that maternal knowledge and sun exposure practices were important determinants of children's vitamin D status (11).

A community-based study conducted in Ghana also reported poor maternal knowledge regarding vitamin D and inappropriate practices related to sunlight exposure among mothers of young children (27). These findings indicate that vitamin D deficiency is increasingly recognized as a public health concern in African countries, particularly in urban areas where lifestyle patterns limit sun exposure.

2.5 Evidence from Ethiopia

In Ethiopia, research related to vitamin D deficiency and maternal awareness remains limited compared to other micronutrient deficiencies such as vitamin A, iron, and iodine. Most national nutrition programs have focused on addressing these micronutrients, while vitamin D has received relatively little attention.

A facility-based cross-sectional study conducted at the University of Gondar Comprehensive Specialized Hospital reported a high prevalence of vitamin D deficiency among pregnant women. The study found that approximately 41% of participants had vitamin D deficiency, indicating a significant public health concern (28).

Another study conducted in Addis Ababa among children attending paediatric clinics also reported the presence of vitamin D deficiency and highlighted the absence of routine vitamin D supplementation programs in Ethiopia (6).

Studies examining maternal practices related to infant sun exposure have also been conducted in some parts of Ethiopia. A community-based study conducted in southwest Ethiopia found that although most mothers had heard about infant sunlight exposure and recognized its benefits, misconceptions and inappropriate practices were still common. The study reported that only about half of the mothers practiced regular infant sun exposure (8).

National surveys such as the Ethiopian Demographic and Health Survey have also documented significant rural-urban differences in maternal education, media exposure, and health service utilization, which may influence maternal knowledge and practices related to child health and nutrition (9; 14).

However, most studies conducted in Ethiopia are facility-based and focus mainly on clinical outcomes rather than maternal knowledge and behaviour. Furthermore, few studies have examined maternal knowledge, attitude, and practice regarding vitamin D in community settings.

2.6 Research Gap Identified

Although several studies have investigated vitamin D deficiency and maternal awareness in different countries, there is limited community-based research in Ethiopia assessing maternal knowledge, attitude and practice regarding vitamin D.

In particular, no study has been conducted in Sekota Town, Sekota Zuria, or wag Himra zone to assess maternal knowledge, attitude, and practice toward vitamin D among mothers of under-five children. Furthermore few studies have compared rural and urban population in relation to maternal awareness and practices regarding vitamin D.

Therefore, this study aims to fill this important research gap by assessing and comparing maternal knowledge, attitude, and practice toward vitamin D among mothers of under-five children in rural and urban areas of Sekota town, and Sekota Zuria.

2.7 conceptual frameworks:

This framework is adapted from previous Ethiopian study, shows factors influencing maternal knowledge, attitude, and practice. Knowledge and attitude are treated as separate dependent variables and also as predictors of practice (29).



Figure 1: Conceptual framework adapted from previous KAP studies showing factors affecting maternal practice toward vitamin D among mothers of under-five children in Sekota Town, and Sekota Zuria Woreda, Amhara Region, Ethiopia, 2026.

OBJECTIVES

3.1 General Objective

To Assess knowledge, Attitude, and Practice Towards Vitamin D Among Mothers of Under-five Children in Rural and Urban Settings of Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026.

3.2 Specific Objectives

1. To assess the level of knowledge about vitamin D among mothers in rural and urban areas of Sekota, and Sekota Zuria, Amhara, Ethiopia, 2026.
2. To determine maternal attitude towards vitamin D among mothers of under-five children in rural and urban areas of Sekota, and Sekota Zuria, Amhara, Ethiopia, 2026.
3. To evaluate maternal practices related to vitamin D among mothers of under-five children in rural and urban areas of Sekota, and Sekota Zuria, Amhara, Ethiopia, 2026.
4. To identify factors associated with maternal knowledge of vitamin D among mothers of under-five children in rural and urban areas of Sekota, and Sekota Zuria, Amhara, Ethiopia, 2026.
5. To identify factors associated with maternal attitude toward vitamin D among mothers of under-five children in rural and urban areas of Sekota, and Sekota Zuria, Amhara, Ethiopia, 2026.
6. To identify factors associated with maternal practice regarding vitamin D among mothers of under-five children in rural and urban areas of Sekota, and Sekota Zuria, Amhara, Ethiopia, 2026.
7. To compare mothers' knowledge, attitudes, and practices regarding vitamin D between rural and urban areas of Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026.

4. METHODS AND MATERIALS

4.1 Study Area and Period

The study was undertaken from Jan 2026 to June 2026, in selected urban and rural areas of Sekota, Wag-Himra zone, Amhara region, northern Ethiopia. Sekota is located 750km north Addis Abeba, according to administrative records and local estimates, Sekota town's population is approximately 41,696 and surrounding rural population is approximately 108,000. There is one general hospital, 7 health centres and 29 health posts (30).

4.2 Study Design

The study were employ community-based comparative cross-sectional study design.

4.3 Population

Source population: the source population was all mothers with under-five children in all urban and rural kebeles of Sekota Town, and Sekota Zuria Woreda.

4.4 Study population

The study population was selected mothers with under-five children in sampled kebeles.

4.5. Inclusion & Exclusion Criteria

4.5.1. Inclusion Criteria

The study were include:

1. Mothers who have at least one under-five children in data collection period.
2. Mothers who have lived in the selected kebeles of Sekota (urban & rural) for at least six months prior to the study.
3. Mothers who are willing and able to provide informed consent and participate in the interview.

4.5.2. Exclusion criteria

The study were exclude:

1. Mothers who are seriously ill during the data collection period and unable to participate in the interview.
2. Mothers who are temporary residents (living in the area of less than six months prior to the data collection period).

4.6 Sample Size

The required sample size of the study participant was determined by using two population proportion formulas (31), since the study compares two groups.

$$n = \frac{(Z_{\alpha/2} + Z_{\beta})^2 [p_1(1-p_1) + p_2(1-p_2)]}{(p_1 - p_2)^2}$$

Where:

n = required sample size for each group

$Z_{\alpha/2}$ = standard normal value at 95% confidence level(1.96)

Z_{β} = standard normal value at 80% power (0.84)

P1 = estimated proportion in first group (urban)

P2 = estimated proportion in second group (rural)

(p1 - p2) = expected difference between the two proportions.

Assume:

Confidence level = 95% → $Z_{\alpha/2} = 1.96$

Power = 80% → $Z_{\beta} = 0.84$

P1 (urban good practice) = 50% (0.50) (32; 33).

P2 (rural good practice) = 35% (0.35) (10; 32)

Margin difference = 0.15

$$n = \frac{(1.96+0.84)^2 [(0.5)(0.5)+(0.35)(0.65)]}{(0.15)^2}$$

n = 166 per group

Total = 332

By adding design effect = 1.5, and 10% non-response rate the final sample size becomes 548 mothers.

4.7 Sampling Procedure

Stratify all 33 kebeles into rural and urban; 9 (30%) kebeles was selected randomly. Then Allocate sample proportionally using Systematic random sampling of households.

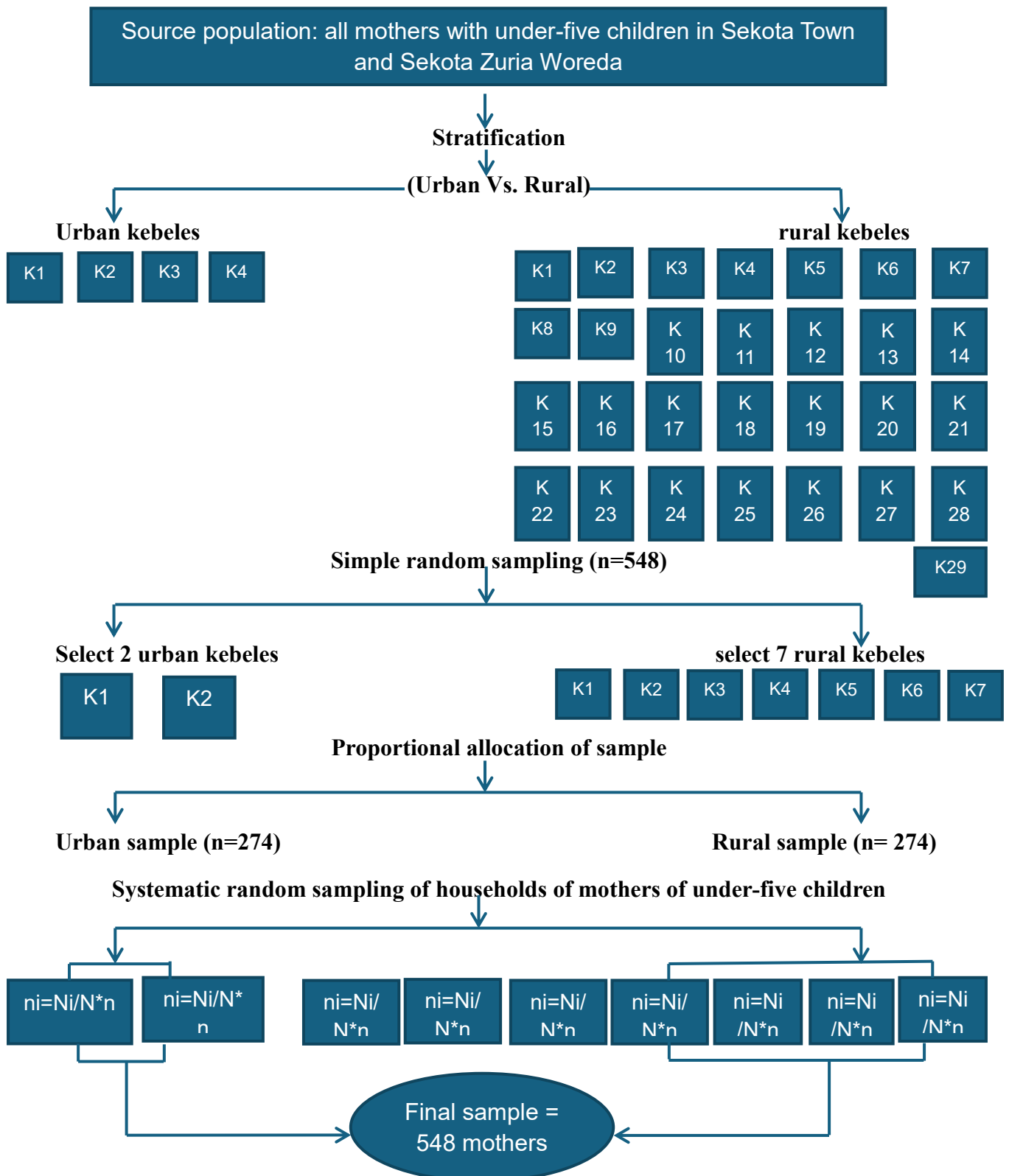


Figure 2: schematic presentation of sampling procedure for the study on maternal knowledge, attitude, and practice toward vitamin D among mothers of under-five children in Sekota Town, and Sekota Zuria Woreda, Amhara region, Ethiopia.

4.8. Variables of the study

4.8.1. Dependent variable

The dependent variable of this study is maternal knowledge, attitude, and practice toward vitamin D.

4.8.2. Independent variable

Table 1: list of independent variables of the study

Socio-demographic variables	Age of mother
	Marital status
	Religion
	Educational status
	Occupation
	Husbands education
	Residence (urban/rural)
Maternal health service variables	Number of under-five children
	ANC attendance
	Postnatal care follow-up
	Health education exposure
Child-related variables	Media exposure
	Age of the youngest child
	Sex of the child
	Birth order of the youngest child
Environmental	House has windows that open
	Child outdoor play area available

4.9. Operational Definitions

Good knowledge: Respondents scoring greater than or equal to the mean score of Knowledge questions ($\geq 7/11$) was categorized as having good knowledge (29).

Positive attitude: respondents scoring greater than or equal to the mean attitude score (≥ 4) was categorized as having positive attitude (34).

Good practice: respondent scoring greater than or equal to the mean practice score ($\geq 6/10$) was categorized as having good practice (29).

4.10 Data Collection procedures and data quality assurance

4.10.1 Questionnaire Development and Validation

The data collection tool was a structured, interviewer-administered questionnaire adapted from validated KAP instruments used in similar Ethiopian studies on sunlight exposure and maternal nutrition. The questionnaire comprises five subscales:

Knowledge (11 items, score 0-11): Awareness of vitamin D, sources (sunlight/diet), benefits, and deficiency consequences.

Attitude (8 items, score 0-8): Perceptions of sunlight importance, supplementation beliefs, and deficiency seriousness (Likert scale: 1=strongly disagree, 2=disagree, 3=neutral, 4=agree, 5=strongly agree).

Practice (10 items, score 0-10): Sun exposure duration/frequency, dietary practices, outdoor habit, and house condition.

Pre-testing: The questionnaire was pre-tested on 40 mothers of under-five children (20 urban, 20 rural) from a similar setting outside the study area, two weeks prior to data collection. Cronbach's alpha was calculated for each subscale (target ≥ 0.70). Items with item-total correlations < 0.30 were revised or removed.

4.10.2 Reliability Measures

Internal consistency: Cronbach's $\alpha \geq 0.70$ per subscale post-pretesting.

Test-retest reliability: 20% of pre-test participants re-interviewed after 7 days (Intra-class correlation coefficient ≥ 0.75).

Inter-rater reliability: Two interviewers were independently score 10% of pre-test interviews (Cohen's $\kappa \geq 0.80$).

4.10.3 Data Quality Control

Supervisors were check 20% of completed questionnaires daily for completeness and consistency.

Double data was entered into EpiData v4.6 with range and consistency checks.

Training: Data collectors (n=6 nurses/HEWs) receive 2-day training on questionnaire administration, informed consent, and cultural sensitivity.

4.11. Data processing and analysis

After data collection, each completed questionnaire was checked daily for completeness, consistency and accuracy by the supervisors and principal investigator. The collected data was coded and entered into EpiData version 4.6 to minimize data entry errors and then exported to SPSS version 26 for analysis.

Before analysis, data cleaning was performed to check for missing values, outliers, and inconsistencies. Descriptive statistics was used to summarize the data. Frequencies, percentages, means and standard deviations was computed for socio-demographic variables and KAP components.

Knowledge, attitude, and practice scores was computed by assigning numerical values to each response, practice related questions was coded based on appropriate and inappropriate behaviours. Mean scores was calculated and respondents scoring greater than or equal to the mean was categorized as having good knowledge, positive attitude and good practice, respectively.

Bivariate Analysis: binary logistic regression was used to identify variables associated with maternal knowledge, attitude, and practice.

Multivariable Analysis: variables with $p < 0.25$ in the bivariate analysis was entered into multivariable logistic regression. Statistical significance was declared at $p < 0.05$ with 95% CI. Hosmer-lemshow test also was used to test model goodness of fit.

To compare rural and urban differences in KAP, Chi-square tests was used. Finally the results was presented using tables, charts and graphs.

4.12 Ethical Consideration

The proposal was reviewed and approved by the institutional review board (IRB) of Woldia University. Permission to conduct the study was also obtained from Sekota Town and Sekota Zuria Woreda.

Informed consent was obtained from each study subject. Each respondent was informed about the objective of the study that it were contribute necessary information for policy maker and other concerned bodies. Any involvement in the study was after their complete consent id obtained. Any study subject who is not wereing to participate in the study were not be forced to participate. They were also informed that all data obtained from them would be kept confidential by using codes instead of any personal identified and is meant only for the purpose of the study.

4.13. Dissemination and utilization of results

The result of the study was presented to the Woldia University, school of public health, and to those significant others who are in need of these results and were request the management of the Woreda health office to act accordingly using the result/findings of the study.

5. RESULTS

5.1 Socio-Demographic Characteristics of the Study Participants

A total of 548 mothers of under-five children participated in this study, yielding a 100% response rate. Mothers aged 25–29 years (29.7%) and 18–24 years (29.2%) constituted the largest age groups. The study included equal proportions of rural and urban mothers (50.0% each). Married mothers accounted for 83.4%, and Orthodox Christians represented 70.3%. Regarding education, 33.0% had primary education, while 22.1% had no formal education. Housewives (32.5%) and farmers (25.5%) were the common occupations. ANC attendance was reported by 64.4%, while PNC attendance was 51.8%. Media exposure was reported by 50.7% of mothers.

Table 2: Socio-Demographic Characteristics of Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Variable	Category	Frequency (n)	Percentage (%)
Age Group (years)	18–24	160	29.2
	25–29	163	29.7
	30–34	128	23.4
	≥35	97	17.7
Residence	Rural	274	50.0
	Urban	274	50.0
Marital Status	Married	457	83.4
	Single	34	6.2
	Divorced	31	5.7
	Widowed	26	4.7
Religion	Orthodox Christian	385	70.3
	Muslim	120	21.9
	Protestant	35	6.4
	Other	8	1.5
Maternal Educational Status	Cannot read and write	121	22.1
	Primary (Grades 1–8)	181	33.0
	Secondary (Grades 9–12)	150	27.4
	Diploma and above	96	17.5
Maternal Occupation	Housewife	178	32.5
	Farmer	140	25.5
	Merchant	106	19.3
	Government employee	93	17.0
	Other	31	5.7
Husband's Educational Status	Cannot read and write	117	21.4
	Primary (Grades 1–8)	170	31.0

	Secondary (Grades 9–12)	156	28.5
	Diploma and above	105	19.2
Monthly Household Income (ETB)	< 2,000	155	28.3
	2,000–5,000	254	46.4
	> 5,000	139	25.4
Sex of Child	Male	278	50.7
	Female	270	49.3
Number of Under-Five Children	One child	158	28.8
	Two children	209	38.1
	Three children	104	19.0
	Four or more children	77	14.1
Age of Youngest Child	<12 months	103	18.8
	12–24 months	109	19.9
	25–59 months	336	61.3
Birth Order of Youngest Child	First child	239	43.6
	Second child	190	34.7
	Third child or above	119	21.7
ANC Attendance	Yes	353	64.4
	No	195	35.6
PNC Attendance	Yes	284	51.8
	No	264	48.2

5.2 Maternal Knowledge about Vitamin D

This section presents the findings on maternal knowledge regarding vitamin D among mothers of under-five children in Sekota Town and Sekota Zuria. Overall, mothers in Sekota Town and Sekota Zuria demonstrated a relatively high level of knowledge regarding vitamin D, with 71.0% (95% CI: 67.2–74.8) classified as having good knowledge. Awareness was particularly strong regarding sunlight as the primary source of vitamin D (85.9%) and its role in bone health (83.4%). However, notable knowledge gaps persisted around specific practical guidance: only 16.8% correctly identified that midday sun exposure carries risk, and merely 15.3% recognized that infants under six months require no direct sun exposure.

Table 3: Distribution of Correct Responses to Knowledge Questions Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Knowledge Questions	Correct (%) n	Incorrect (%) n
Vitamin D is essential for bone health and prevents rickets	457 (83.4)	91 (16.6)

Sunlight is the primary vitamin D source	471 (85.9)	77 (14.1)
Midday sun is dangerous for safe exposure	92 (16.8)	456 (83.2)
Dark skin needs more sun exposure than light skin for vitamin D	406 (74.1)	142 (25.9)
Covering a child's entire body prevents vitamin D deficiency	404 (73.7)	144 (26.3)
10-30 minutes of morning sunlight daily is enough for infants	439 (80.1)	109 (19.9)
Foods alone are sufficient without sunlight exposure	399 (72.8)	149 (27.2)
Cloudy weather prevents all vitamin D production from sun	382 (69.7)	166 (30.3)
Vitamin D deficiency causes only weak bones	379 (69.2)	169 (30.8)
Infant under 6 months needs no direct sun exposure	84 (15.3)	464 (84.7)
Air pollution greatly reduces skin's vitamin D synthesis	411 (75.0)	137 (25.0)

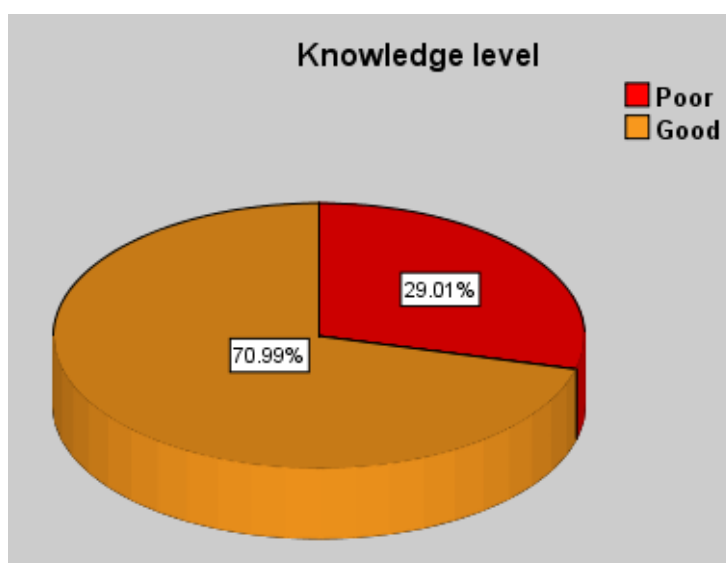


Figure 3: Level of Knowledge Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548).

5.3 Maternal Attitude toward Vitamin D

This section presents the findings on maternal attitudes toward vitamin D among mothers of under-five children in Sekota Town and Sekota Zuria. More than half of mothers (58.6%) held

a favorable attitude toward vitamin D, though the margin over unfavorable attitude was moderate rather than pronounced. Responses across individual items were dominated by neutral positions (34.7–41.6% for most statements), indicating widespread ambivalence rather than firm conviction in either direction. Agreement was comparatively higher for practical, action-oriented statements such as scheduling outdoor time (35.6%) and midday sun exposure (37.8%), while misconception-related statements supplement unnecessary (23.7%) and blanket sun protection (25.0%) drew the lowest agreement.

Table 4: Attitude Responses Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Attitude Statements	Strongly Disagree (%)	Disagree n (%)	Neutral n (%)	Agree n (%)	Strongly Agree n (%)
I believe morning sun benefits child health	14 (2.6)	113 (20.6)	220 (40.1)	148 (27.0)	53 (9.7)
Sunlight increases skin cancer risk more than deficiency risk	35 (6.4)	153 (27.9)	212 (38.7)	118 (21.5)	30 (5.5)
Worth scheduling daily outdoor time for child vitamin D	31 (5.7)	132 (24.1)	190 (34.7)	149 (27.2)	46 (8.4)
Midday sun is best for quick vitamin D in children	36 (6.6)	113 (20.6)	192 (35.0)	167 (30.5)	40 (7.3)
Vitamin D supplements are unnecessary with enough sun	52 (9.5)	153 (27.9)	213 (38.9)	105 (19.2)	25 (4.6)
Protecting kids from all sun is better than risking rickets	45 (8.2)	158 (28.8)	208 (38.0)	107 (19.5)	30 (5.5)
My family tradition supports infant sun exposure	34 (6.2)	119 (21.7)	207 (37.8)	137 (25.0)	51 (9.3)
Need community health education on vitamin D	34 (6.2)	108 (19.7)	228 (41.6)	140 (25.5)	38 (6.9)

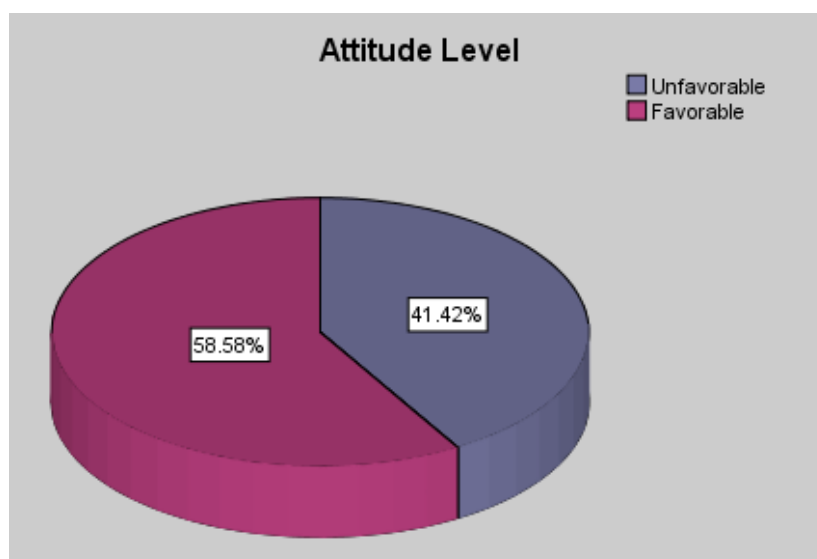


Figure 4: Level of Attitude Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548).

5.4 Maternal Practice towards Vitamin D

The study found that 48.4% of mothers had good practice regarding vitamin D, while 51.6% had poor practice, with a mean practice score of 5.40 out of 10. While many reported children consuming vitamin D-rich foods (58.0%) and adequate outdoor playtime (43.8%), actual sun exposure frequency was low, with 53.3% rarely exposing their child. Protective behaviors were mixed, as nearly half applied heavy covering (46.4%) or sunscreen/oils (49.8%), both reducing synthesis. Only 43.1% opened windows despite 68.8% having them, showing favorable conditions don't consistently translate into optimal practice.

Table 5: Practice Responses Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Practice Item	Response	n (%)
Exposes child to sunlight	Yes	291 (53.1)
	No	257 (46.9)
Days per week exposed to sunlight	Daily	116 (21.2)
	3–4 days	92 (16.8)
	1–2 days	48 (8.8)
	Rarely	292 (53.3)
Time of sunlight exposure	Morning (8–10 AM)	244 (44.5)
	Midday	190 (34.7)
	Evening	114 (20.8)
Duration of sunlight exposure	10–30 minutes	248 (45.3)
	<10 minutes	170 (31.0)
	>30 minutes	130 (23.7)
Child wears heavy covering during exposure	No	294 (53.6)

	Yes	254 (46.4)
Uses sunscreen/skin oils during exposure	No	275 (50.2)
	Yes	273 (49.8)
Weekly outdoor playtime	>30 minutes/day	240 (43.8)
	10–30 minutes/day	188 (34.3)
	<30 minutes/day	120 (21.9)
Consumes vitamin D-rich foods (eggs, fish)	Yes	318 (58.0)
	No	230 (42.0)
House has windows	Yes	377 (68.8)
	No	171 (31.2)
Opens windows for sunlight at home	Yes	236 (43.1)
	No	312 (56.9)

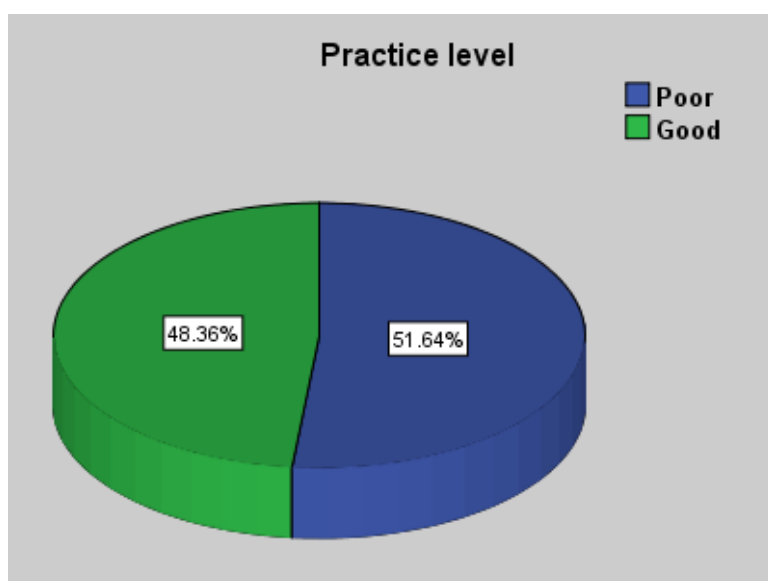


Figure 5: Level of Practice Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548).

5.5 Comparison of Knowledge, Attitude, and Practice between Rural and Urban Mothers

This section compares maternal knowledge, attitude, and practice regarding vitamin D between rural and urban mothers in Sekota. A chi-square test was used to assess statistical significance. Urban mothers had significantly higher good knowledge (86.5% vs 55.5%, $p<0.001$) and good practice (54.0% vs 42.7%, $p=0.008$) compared to rural mothers. Although favorable attitude was higher among urban mothers (62.4% vs 54.7%), the difference was not statistically significant ($p=0.069$).

Table 6: Comparison of KAP Levels by Residence Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Variables	Rural (n=274) n (%)	Urban (n=274) n (%)	Total (n=548) n (%)	χ^2	p-value
Knowledge Level				64.014	<0.001
Poor	122 (44.5)	37 (13.5)	159 (29.0)		
Good	152 (55.5)	237 (86.5)	389 (71.0)		
Attitude Level				3.317	0.069
Unfavorable	124 (45.3)	103 (37.6)	227 (41.4)		
Favorable	150 (54.7)	171 (62.4)	321 (58.6)		
Practice Level				7.022	0.008
Poor	157 (57.3)	126 (46.0)	283 (51.6)		
Good	117 (42.7)	148 (54.0)	265 (48.4)		

5.6 Factors Associated with Maternal Knowledge, Attitude, and Practice

5.6.1 Factors Associated with Maternal Knowledge of Vitamin D

The multivariable analysis identified five independent predictors of good knowledge about vitamin D. Urban mothers were 2.05 times more likely to have good knowledge than rural mothers (AOR=2.05, p=0.008). Maternal education showed a dose-response relationship, with secondary education (AOR=6.22, p<0.001) and diploma+ (AOR=5.89, p<0.001) being the strongest predictors. ANC attendance (AOR=1.92, p=0.007), media exposure (AOR=1.84, p=0.009), and favorable attitudes (AOR=2.39, p<0.001) were also significantly associated with good knowledge

Table 7: Factors Associated with Maternal Knowledge of Vitamin D Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Variables	Category	Knowledge Level		COR (95% CI)	AOR (95% CI)	p-value
		Good	Poor			
Residence	Rural	152	122	1	1	
	Urban	237	37	5.14 (3.38–7.83)	2.05 (1.20–3.49)	0.008
Maternal Education	Cannot read & write	39	82	1	1	
	Primary (1–8)	133	48	1.89 (1.19–3.02)	1.85 (1.10–3.11)	0.020
	Secondary (9–12)	129	21	6.67 (3.73–11.95)	6.22 (3.27–11.85)	<0.001
	Diploma+	88	8	10.50 (4.85–22.76)	5.89 (2.52–13.78)	<0.001
	No	112	83	1	1	

ANC Attendance	Yes	277	76	3.39 (2.31–4.99)	1.92 (1.20–3.07)	0.007
Media Exposure	No	156	114	1	1	
	Yes	233	45	2.85 (1.94–4.20)	1.84 (1.17–2.92)	0.009
Attitude Level	Unfavorable	119	108	1	1	
	Favorable	270	51	2.49 (1.71–3.63)	2.39 (1.54–3.69)	<0.001

5.6.2 Factors Associated with Maternal Attitude Toward Vitamin D

The multivariable analysis identified knowledge, practice, and PNC attendance as independent predictors of favorable attitude. Mothers with good knowledge (AOR=2.47, $p<0.001$) and good practice (AOR=1.64, $p=0.007$) were significantly more likely to have favorable attitudes. However, PNC attendance was inversely associated with favorable attitude (AOR=0.58, $p=0.005$), suggesting that PNC contacts may increase awareness of sun exposure risks, leading to more cautious attitudes. ANC attendance was not significant ($p=0.062$).

Table 8: Factors Associated with Maternal Attitude Toward Vitamin D Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Variables	Category	Attitude level		COR (95% CI)	AOR (95% CI)	p-value
		Favorable	Unfavorable			
ANC Attendance	No	100	95	1	1	
	Yes	221	132	1.59 (1.12–2.27)	1.46 (0.98–2.17)	0.062
PNC Attendance	No	163	101	1	1	
	Yes	158	126	0.78 (0.55–1.09)	0.58 (0.40–0.84)	0.005
Knowledge Level	Poor	68	91	1	1	
	Good	253	136	2.49 (1.71–3.63)	2.47 (1.65–3.70)	<0.001
Practice Level	Poor	148	135	1	1	
	Good	173	92	1.72 (1.22–2.42)	1.64 (1.15–2.35)	0.007

COR = Crude Odds Ratio; AOR = Adjusted Odds Ratio; CI = Confidence Interval; Ref = Reference Category

5.6.3 Factors Associated with Maternal Practice Regarding Vitamin D

The multivariable analysis identified residence and attitude as independent predictors of good practice. Urban mothers were 1.65 times more likely to have good practice than rural mothers

(AOR=1.65, p=0.040). Mothers with favorable attitudes were 1.66 times more likely to have good practice (AOR=1.66, p=0.005). Occupation and ANC attendance were not significantly associated with practice in the multivariable model.

Table 9: Factors Associated with Maternal Practice Regarding Vitamin D Among Mothers of Under-Five Children in Sekota Town and Sekota Zuria, Amhara Region, Ethiopia, 2026 (n=548)

Variables	Category	Practice Level		COR (95% CI)	AOR (95% CI)	p-value
		Good	Poor			
Residence	Rural	117	157	1	1	
	Urban	148	126	1.58 (1.13–2.21)	1.65 (1.02–2.67)	0.040
Occupation	Farmer	64	76	1	1	
	Government employee	47	46	1.21 (0.72–2.05)	0.63 (0.32–1.22)	0.171
	Housewife	82	96	1.01 (0.65–1.58)	0.69 (0.41–1.14)	0.144
	Merchant	49	57	1.02 (0.62–1.69)	0.56 (0.30–1.04)	0.067
	Other	23	8	3.41 (1.43–8.15)	2.24 (0.90–5.60)	0.083
ANC Attendance	No	78	117	1	1	
	Yes	187	166	1.69 (1.19–2.41)	1.45 (0.96–2.19)	0.080
Attitude Level	Unfavorable	92	135	1	1	
	Favorable	173	148	1.72 (1.22–2.42)	1.66 (1.17–2.37)	0.005

6. DISCUSSION

This community-based comparative cross-sectional study assessed knowledge, attitude, and practice (KAP) toward vitamin D among 548 mothers of under-five children in rural and urban kebeles of Sekota Town and Sekota Zuria, Amhara Region, Ethiopia. Overall, 71.0% of mothers had good knowledge, 58.6% held a favorable attitude, and only 48.4% demonstrated good practice. This declining gradient across the three domains—knowledge exceeding attitude, and attitude exceeding practice—is consistent with the classic KAP gap reported extensively in the international and African literature, and forms the central narrative of this discussion: awareness of vitamin D exists, but it does not reliably translate into protective behavior.

The finding that 71.0% of mothers had good knowledge situates Sekota within the upper-middle range of knowledge levels reported across comparable settings. This result is considerably more favorable than reports from India, where a community-based study found that approximately two-thirds of mothers had poor knowledge about vitamin D (19), and another Indian study similarly reported that more than 60% of mothers lacked adequate knowledge (20). It is also more favorable than findings from Nigeria and Ghana, where maternal knowledge regarding vitamin D was generally described as inadequate (26; 27). However, it is broadly comparable to the Saudi Arabian context, where nearly 70% of mothers were reported to be aware of vitamin D (22), and to the southwest Ethiopian study, in which most mothers had heard of infant sunlight exposure and recognized its benefits (8).

Item-level analysis, however, reveals that this seemingly favorable overall knowledge score masks important gaps in practically relevant knowledge. While awareness of sunlight as the primary vitamin D source (85.9%) and its role in bone health (83.4%) was high—mirroring the general-knowledge strength seen in the Malaysian and Saudi studies (22; 23)—mothers performed poorly on safety-specific items: only 16.8% correctly identified that midday sun exposure carries risk, and only 15.3% knew that infants under six months require no direct sun exposure. This pattern, in which general conceptual knowledge outpaces specific procedural knowledge, echoes the Malaysian finding that higher maternal education was associated with better knowledge of sources and benefits, but not necessarily with correct application of that knowledge in daily practice (23). It suggests that health education messaging in Sekota has successfully conveyed the *existence* and *importance* of vitamin D, but has not adequately equipped mothers with the specific behavioral guidance needed to translate that awareness into safe practice.

Just under three-fifths of mothers (58.6%) held a favorable attitude toward vitamin D. This finding is comparable to attitude levels reported in other Ethiopian facility-based KAP studies, which similarly identified attitude as a weaker or more ambivalent domain relative to knowledge. The dominance of neutral responses across nearly every attitude item (34.7–41.6%) is a particularly notable finding, indicating that many mothers have not yet formed firm convictions whether positive or negative regarding vitamin D-related behaviors. This ambivalence is consistent with the broader African evidence suggesting that awareness alone does not necessarily produce confident, actionable beliefs (10; 11), and it may partly explain why attitude, unlike knowledge, showed no significant urban-rural difference ($p=0.069$) in this study: attitudes may be shaped less by residence-driven exposure to information and more by deeper cultural or experiential factors that do not differ systematically between rural and urban populations in this particular setting.

Practice was the weakest of the three domains, with only 48.4% of mothers demonstrating good practice – a finding that reinforces the well-documented disconnect between knowledge and behavior seen throughout this literature. This result is comparable to the southwest Ethiopian study, where only about half of mothers practiced regular infant sun exposure (8), and to the Turkish study, in which only about 30% of mothers practiced appropriate sun exposure timing despite widespread awareness (25). It is considerably lower than the near-universal exposure recognition reported in some other settings but aligns closely with the Kenyan comparative study, which similarly found that awareness did not guarantee appropriate sun exposure behavior (10).

The item-level practice data further illuminate this gap: although a majority of mothers reported feeding their children vitamin D-rich foods (58.0%) and providing adequate outdoor playtime (43.8%), actual sunlight exposure frequency was low, with 53.3% of mothers rarely exposing their child to sunlight. Protective behaviors that inadvertently block cutaneous synthesis – heavy covering (46.4%) and sunscreen or oil application (49.8%) – were also common, suggesting that protective instincts, likely rooted in the same cultural avoidance of sun exposure documented among Saudi mothers (21), may be actively undermining vitamin D synthesis even among mothers who intend to care for their children well. Similarly, despite 68.8% of households having windows, only 43.1% of mothers opened them for sunlight, indicating that structural or environmental opportunity does not automatically translate into behavioral practice. Taken together, these findings suggest that improving practice in this setting requires more than raising awareness; it requires directly addressing culturally rooted protective behaviors that, while well-intentioned, are counterproductive.

The comparative analysis revealed that urban mothers had significantly better knowledge (86.5% vs. 55.5%, $p<0.001$) and better practice (54.0% vs. 42.7%, $p=0.008$) than their rural counterparts, while attitude did not differ significantly by residence ($p=0.069$). This urban advantage in knowledge is consistent with the general pattern observed in the Kenyan and South African studies, where urban mothers demonstrated superior knowledge attributable to greater access to media, health services, and formal education (10; 11). Ethiopian Demographic and Health Survey data similarly document substantial rural-urban disparities in education, media exposure, and health service utilization that plausibly explain this gradient (9; 14).

Notably, the urban practice advantage found in this study contrasts with the Kenyan comparative study, which reported that urban mothers had *better knowledge but poorer* sun exposure practices than rural mothers, attributing this reversal to indoor urban lifestyles (10) a pattern also echoed in the Malaysian study (23). The fact that Sekota's urban mothers outperformed rural mothers on *both* knowledge and practice suggests that in this specific setting, urban residence confers practical advantages (e.g., greater household resources, better health service access, more consistent ANC/PNC contact) that outweigh any indoor-lifestyle penalty observed elsewhere. This divergence highlights that the relationship between urbanicity and vitamin D-related practice is not universal but is shaped by local infrastructure, health system contact, and lifestyle patterns specific to each study context.

The multivariable analysis identified a coherent set of determinants that collectively explain the observed KAP patterns. Maternal education emerged as the strongest predictor of good knowledge, with a clear dose-response gradient (secondary education AOR=6.22; diploma and above AOR=5.89, both $p<0.001$), reaffirming the education-knowledge relationship consistently reported across the Malaysian, Turkish, and broader African literature (23; 25; 27). Urban residence (AOR=2.05, $p=0.008$), ANC attendance (AOR=1.92, $p=0.007$), and media exposure (AOR=1.84, $p=0.009$) further independently predicted good knowledge, echoing the Turkish finding that antenatal counselling was significantly associated with improved knowledge (25) and reinforcing EDHS-documented rural-urban gaps in media and health service access (9; 14).

The bidirectional relationship between knowledge, attitude, and practice observed in this study is theoretically important. Favorable attitude was independently associated with good knowledge (AOR=2.39, $p<0.001$) and, in turn, good knowledge (AOR=2.47, $p<0.001$) and good practice (AOR=1.64, $p=0.007$) were independently associated with favorable attitude, while favorable attitude also predicted good practice (AOR=1.66, $p=0.005$). This mutually reinforcing structure is consistent with the conceptual framework adopted for this study, in

which knowledge and attitude function simultaneously as determinants and outcomes within the KAP model (29), rather than as a simple unidirectional pathway.

One counterintuitive finding warrants particular attention: PNC attendance was *inversely* associated with favorable attitude (AOR=0.58, $p=0.005$). A plausible explanation, consistent with the cautionary counseling patterns noted in the Saudi and Middle Eastern literature (21), is that postnatal contact with health workers may increase mothers' awareness of potential risks associated with sun exposure (e.g., skin damage, overheating), fostering a more cautious rather than uniformly favorable attitude. This finding suggests that health messaging delivered during PNC visits may inadvertently emphasize risk avoidance over the benefits of appropriate, moderate sun exposure—an important consideration for refining counseling content.

Finally, urban residence (AOR=1.65, $p=0.040$) and favorable attitude (AOR=1.66, $p=0.005$) independently predicted good practice, while occupation and ANC attendance lost significance in the adjusted model. This indicates that once attitude and residence are accounted for, occupational category does not independently drive practice, suggesting that interventions targeting attitude change may have broader reach across occupational groups than occupation-specific programming would.

7. STRENGTHS AND LIMITATIONS OF THE STUDY

7.1 Strengths

- Community-based design ensured representation of both rural and urban populations, minimizing selection bias
- Comparative rural-urban design provided valuable contextual insights for targeted interventions
- Validated and pre-tested questionnaire with good internal consistency (Cronbach's $\alpha \geq 0.70$)
- High response rate (100%) strengthened generalizability and minimized non-response bias

7.2 Limitations

- Cross-sectional design limits causal inference; only associations can be identified
- Self-reported data subject to social desirability bias and overreporting of positive behaviors
- Recall bias may affect accuracy of sun exposure frequency and duration reporting
- Limited generalizability beyond Sekota Town and Sekota Zuria

8. CONCLUSION AND RECOMMENDATIONS

8.1 Conclusion

This study assessed maternal knowledge, attitude, and practice regarding vitamin D among mothers of under-five children in Sekota Town and Sekota Zuria. The findings revealed that the majority of mothers had good knowledge and favorable attitudes toward vitamin D. However, maternal practices remained suboptimal, with a notable gap between knowledge and actual practice. Significant urban-rural disparities were observed, with urban mothers demonstrating better knowledge and practices compared to rural mothers. Residence, maternal education, ANC attendance, media exposure, attitude, and PNC attendance were identified as significant predictors of KAP. The study highlights the need for targeted health education interventions, particularly focusing on safe sun exposure practices and addressing misconceptions, to bridge the knowledge-practice gap and improve vitamin D status among children in the study area.

8.2 Recommendations

For Health Education and Promotion: Develop and implement targeted health education programs focusing on safe sun exposure timing and infant sun exposure practices through ANC, PNC, and immunization services.

For Media and Communication: Utilize mass media (radio, TV) to disseminate vitamin D education, particularly targeting rural mothers with limited access to health facilities.

For Health System Strengthening: Integrate vitamin D education into routine maternal and child health services and train health workers on counseling mothers about vitamin D.

For Community Engagement: Engage community health workers (Health Extension Workers) to conduct home visits and community conversations addressing misconceptions about sun exposure.

For Future Research: Conduct biochemical assessments of vitamin D status, qualitative studies, intervention trials, and national-level surveys to inform policy and program development.

9. REFERENCE

1. The vitamin D deficiency pandemic. Holick, Micheal F. 2017, *Rev Endocr Metab Disord*, pp. 18, 153-165.
2. Is vitamin D deficiency a major global public health problem? Cristina Palacios, Lillina Gonzalez. 2014, *J Steroid Biochem Mol Disord.*, pp. 138-145.
3. Vitamin D deficiency. Holick, Micheal F. 2007, *N Engl J Med*, pp. 266-281.
4. Dietary reference intakes for calcium and vitamin D. A. Catharine Ross, et al. 2011, *J Clin Endocrinol Metab.*, pp. 53-58.
5. Vitamin D deficiency in children and adolescents. Mira, Madhusmita. 2008, *pediatrics*, pp. 398-417.
6. vitamin D deficiency among children in Addis Abeba. Teshome, Biniyam. 2017, *Ethio J health Sci*, pp. 27: 473-480.
7. Global overview of vitamin D status. Natasja M. van Schoor, Paul Lips. 2017, *Endocrinol Metab clin North Am*, pp. 845-870.
8. Knowledge and practice of infant exposure to sunlight among mothers in rural villages southwest, Ethiopia. Tadesse A, Yeshanew S and Gedefa AG. mettu district : *front. public health*, 2022, *Ethip J Health Dev.*, pp. 36: 112-120. doi: 10.3389/fpubh.2023.1166976.
9. Central Statistical Agency, ICF. Ethiopian Mini Demographic and Health Survey 2019. Addis Abeba, Ethiopia. : Central Statistical Agency, ICF, 2019.
10. prevalence of vitamin D deficiency in africa: a systematic review and meta-analysis. Mogire, Robert M. 2020;8, *Lancet Glob Health*, pp. 134-142.
11. vitamin D status among South Africa children. Poopedi, Mary A. 2011;14, *public health Nutr.*, pp. 234-239.
12. World Health Organization. Micronutrient deficiencies: global health issues. Geneva : WHO, 2019.
13. Global Vitamin D Status in Children: a systematic review. Cashman, Kevin D. 2023, *Nutrients*, p. 15;1987.
14. United Nations. Transforming our world: the 2030 Agenda for sustainable Development. New York : United Nations, 2015.
15. Federal Ministry of Health Ethiopia. Health Sector Transformation Plan II (HSTP II): 2020/21-2024/25. Addis Abeba. : FMOH, 2021.
16. prevalence and determinants of vitamin D deficiency among pregnant women in Gondar town Ethiopia. Chane, Elias. 2024, *BMJ open*, pp. 5-10.

17. Maternal Knowledge and Practices Regarding Vitamin D in Turkey. Yilmaz, Recep. 2024, BMC Public Health., p. 15;1987.
18. FDRE, Federal Ministry of Health. National Nutritional Program II (2016-2020). Addis Abeba : FMOH, 2016.
19. Federal Democratic Republic of Ethiopia. Food and Nutrition Policy. Addis Abeba : Government of Ethiopia., 2018.
20. Link Nutrition Causal Analysis. Final Report: Nutrition causal analysis in Sekota and Dehana Woredas, Wag-Himra Zone. Wag-Himra : Link NCA, 2018.
21. An Experimental Investigation on rickets . Mellanby, Edward. 1921, lancet, pp. 407-412.
22. studies on experimental rickets. Mccollum, Elmer V. 1922, J Biol Chem, pp. 293-312.
23. Vitamin D and rickets: historical perspective. Rajakumar, Kumaravel. 2003, pediatrics, pp. 132-135.
24. serum 25-hydroxyvitamin D status in the US population:NHANES 2001-2004. Looker, Anne C. 2008, Am J Clin Nutr, pp. 1519-1527.
25. A Systematic review of vitamin D status in populations worldwide. Hilger, Jans. 2014, Br J Nutr., pp. 111: 23-45.
26. vitamin D status in India: its implication and health consequences. Marwaha, R. K. 2010, India J Med Res, pp. 636-656.
27. vitamin D status in india. C. V. Harinarayan, Sanjay R. Joshi. 2009, J Assoc Physicians India, pp. 40-48.
28. vitamin D status among Saudi populations. Ardawi, Mohammed S. M. 2012;23, Osteoporos int, pp. 613-621.
29. Ayele FY, Kebede N, Mkonen AM, Mihiretu MM, Tsega Y, Addisu E, et al. sunlight exposure practice and its associated factors among infants in Ethiopia systmatic review and meta analysis. Addis Abeba : PLoS ONE, 2024. 19(11).
30. Central Statistical Agency, ICF. Ethiopian Demographic and Health Survey2016. Addis Abeba, Ethiopia. : Central Statistical Agency, ICF, 2016.
31. LemeshowS, Hosmer DW Jr, Klar J, Lwanga SK. Adequacy of Sample Size in Health Studies. Geneva : World Health Organization, 1990.
32. Knowledge, Attitude, and Practice of Vitamin D Intake Among Pregnant Women... al, Eyong E et. calabar, Nigeria : IBOM MEDICAL JOURNAL, 2025, Niger J Clin Pract., Vol. Vol.18 No.3, pp. 760-766. 10.61386/imj.v18i3.700.
33. MBC Pediatr, p. 23;487.

34. Attitude of mothers attending public hospital in Addis Abeba. Yohannes Godie, Girum Teshome, Debela Gela, Mekonen Adimasu. 1, Addis Abeba : BMJ Paediatrics Open, 2022, Vol. 6.
35. vitamin D status among saudi mothers and childrens. Alfawaz, Huda. 2014, MBC Public health, p. 1291.
36. Vitamin D Deficiency in Africa: burden and predictors. Mogrie, Robert M. 2021, Nutrients, p. 13;3727.
37. vitamin D status among Malaysian mothers and children. Poh., Bee Koo. 2016;19, public health Nutr, pp. 1393-1402.
38. Tefera, Haileyesus Gedamu and Ylikal. Assessment of knowkedge, attitude, and practice of sunlight exposure of infants among mothers attending governmental health facilities in farta district, south Gondar zone,Ethiopia. Farta District, South Gondar Zone. : International Jornal of Reproductive Medicine, 2018.

10. ANNEXS

Annex-I Information Sheet to get Permission for the Research

Introduction

This information sheet is prepared to explain the research project that you are asked to join by a group of research investigators. The main aim of this research project is to assess Knowledge, Attitude, and Practice of under-five mothers Toward Vitamin D, in urban and rural kebeles of sekota, Amhara region, Ethiopia, 2026. The research team includes a final year MPH graduate student in human nutrition and two senior advisors from Woldia University, institute of Public health, college of medicine and health sciences in the university.

Name of Principal Investigator: Fikadu Tazeze

Name of Advisors: Dr. Melese Abate Reta (PhD)

Mr. Moges Milashu (MPH)

Name of the Sponsor: Woldia University

Name of Organization: Woldia University, College of Medicine and Health Sciences, Institute of Public Health:

This information sheet is prepared by above mentioned researcher whose main aim

Is to assess Knowledge, Attitude, and Practice of under-five mothers Toward Vitamin D, in urban and rural kebeles of sekota, Amhara region, Ethiopia, 2026. The investigator is a final year MPH student with advisors from institute of public health, college of medicine and health sciences, university of Gondar.

Purpose:

The purpose of this research study is to assess Knowledge, Attitude, and Practice of under-five mothers Toward Vitamin D, in urban and rural kebeles of sekota, Amhara region, Ethiopia, 2026. Results from this study was used to assist in making recommendations for those who are responsible and in need of the information.

Procedure:

This study uses community based cross-sectional study design and the respondent's information was collected using structured questionnaire via interview. Ethical clearance was

found from Woldia University. The study involves mothers of under-five children during the data collection period in rural and urban kebeles of sekota.

Risk and/or Discomfort:

There is no risk or discomfort that you were face by participating in this research except dedication of time for participating in the study. Any personal information registered were not be copied and transferred to other bodies. Every piece of information was kept confidentially. There is no risk of participation in this research project.

Benefits:

You may not receive a direct benefit from participating in this study. However , the information gathered may help improve maternal and child health programs in your community.

Incentives/Payment for Participating:

There is no incentive or payment to be gained by taking part in this project.

Confidentiality:

All Personal identifiers & personal information were not be taken. The information

Collected from this research project was kept confidential. Information was accessed by researcher and research assistant only.

Persons to contact:

This research project was reviewed and approved by the ethical committee of the Woldia University. If you want to know more information you can contact the committee through the address below. If you have any question you may contact the Following individuals.

Investigator: Fikadu Tazeze

Advisors: Dr. Melese Abate (PhD)

Mr. Moges Milashu (MPH)

Annex II Declaration

I, the undersigned, public health nutrition student declare that this proposal is my original work in partial fulfilment of the requirement for the degree of Master of Public Health in Human nutrition.

Name: Fikadu Tazeze

Signature: _____ Place of submission: Institute of Public Health, College of Medicine and Health Sciences, Woldia University.

Date of Submission: _____

This proposal work has been submitted for examination with our approval as university advisors.

Advisors Name & Signature

1. Dr. Melese Abate (PhD) _____

2. Mr. Moges Milashu (MPH) _____

Annex III. Assurance of investigator

The undersigned agrees to accept responsibility for the scientific, ethical and technical

Conduct of the research project and for provision of required progress reports as preterm and conditions of the Institutional Review Board of the Woldia University.

Name of the student: Fikadu Tazeze

Date: _____ Signature: _____

Approval of the advisors

Advisors

Name Signature Date

1. Dr. Melese Abate (PhD) _____

2. Mr. Moges Milashu (MPH) _____

Annex VI – questionnaire

Title: Assessment of KAP toward vitamin D among mothers of under-five children in rural and urban setting of sekota, comparative study.

Informed Consent Form

Purpose: You are invited to participate in a research study conducted by an MPH student from Woldia University. The purpose is to assess knowledge, attitude, and practice toward vitamin D.

Procedure: You was asked questions about vitamin D, sun exposure, and child feeding practices. The interview were take approximately 20 minutes.

Confidentiality: Your information were remain confidential. No names was recorded

Voluntary Participation: Participation is voluntary. You may refuse or withdraw at any time.

Are you wereing to participate in the study?

Yes No

If you are wereing to participate in this study please go to the next page.

For any further question, contact the investigator:

Name of investigator: Fikadu Tazeze

Telephone number: 0915618563

Consent Statement: I have read and understood the information above and agree to participate.

Name: _____

Signature: _____

Date: _____

SECTION I: IDENTIFICATION

1. Questionnaire Code: _____
2. Date of Interview: ____ / ____ / 2026
3. Residence: ☐ 1 = Urban ☐ 2 = Rural
4. Kebele: _____
5. Interviewer ID: _____

S.NO	SECTION II: SOCIO-DEMOGRAPHIC CHARACTERISTICS	
	Question's	Answers
1	Age of mother (years)	-----yr
2	Marital status	☐ Married ☐ Single ☐ Divorced ☐ Widowed
3	Religion	☐ Orthodox ☐ Muslim ☐ Protestant ☐ Other _____
4	Educational status	☐ Cannot read & write ☐ Primary (1–8) ☐ Secondary (9–12) ☐ Diploma+
5	Occupation	☐ Housewife ☐ Government employee ☐ Merchant ☐ Farmer ☐ Other _____
6	Husband education level:	☐ No formal education ☐ Primary ☐ Secondary ☐ Diploma+
7	Monthly household income (ETB)	☐ <2000 ☐ 2000–5000 ☐ >5000
8	Number of under-five children.	-----
9	Age of youngest child (months)	-----month
10	Sex of the child	☐ Male ☐ Female
11	Birth order of youngest child	☐ 1st ☐ >3rd ☐ 2 nd
12	ANC attendance during last pregnancy	☐ Yes ☐ No
13	PNC attendance during last delivery	☐ Yes ☐ No
14	Media exposure (Radio/TV/Health education)	☐ Yes ☐ No

	SECTION III: KNOWLEDGE ABOUT VITAMIN D				
1	Vitamin D is essential for bone health and prevents rickets.	☐ Yes	☐ No	☐ Don't know	
2	Sunlight is the primary vitamin D source.	☐ Yes	☐ No	☐ Don't know	
3	Midday sun is best for safe exposure.	☐ Yes	☐ No	☐ Don't know	
4	Dark skin needs more sun exposure than light skin for vitamin D.	☐ Yes	☐ No	☐ Don't know	
5	Covering a child's entire body prevents vitamin D deficiency.	☐ Yes	☐ No	☐ Don't know	
6	10-30 min of morning sunlight daily is enough for infants.	☐ Yes	☐ No	☐ Don't know	
7	Foods alone are sufficient without sunlight exposure.	☐ Yes	☐ No	☐ Don't know	
8	Cloudy weather prevents all vitamin D production from sun.	☐ Yes	☐ No	☐ Don't know	
9	Vitamin D deficiency causes only weak bones.	☐ Yes	☐ No	☐ Don't know	
10	Infant under 6 months needs no direct sun exposure.	☐ Yes	☐ No	☐ Don't know	
11	Air pollution greatly reduces skin's vitamin D synthesis	☐ Yes	☐ No	☐ Don't know	
	SECTION IV: ATTITUDE TOWARDS VITAMIN D				
		1= strongly disagree, 2= Disagree, 3= Neutral, 4=Agree, 5=Strongly Agree			
1	I believe morning sun benefits child health.	1 ☐	2 ☐	3 ☐	4 ☐ 5 ☐
2	Sunlight for babies increases skin cancer risk than deficiency risk.	1 ☐	2 ☐	3 ☐	4 ☐ 5 ☐
3	It's worth scheduling daily outdoor time for my child vitamin D.	1 ☐	2 ☐	3 ☐	4 ☐ 5 ☐
4	Midday sun is best for quick vitamin D in children	1 ☐	2 ☐	3 ☐	4 ☐ 5 ☐

5	Vitamin D supplements are unnecessary with enough sun.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
6	Protecting kids from all sun is better than risking rickets.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
7	My family tradition support infant sun exposure.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
8	need community health education on vitamin	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
SECTION V: PRACTICE RELATED TO VITAMIN D		
1	Do you expose your child to sunlight?	☐ Yes ☐ No
2	If yes Q1, for How many days per week?	☐ Daily ☐ 3–4 days ☐ 1–2 days ☐ Rarely
3	Time of exposure:	☐ Morning (8–10 AM) ☐ Midday ☐ Evening
4	Duration of exposure:	☐ <10 minutes ☐ 10–30 minutes ☐ >30 minutes
5	Does the child wear heavy covering during exposure?	☐ Yes ☐ No
6	Do you use sunscreen or other skin oils during morning sun exposure?	☐ Yes ☐ No
7	Childs weekly outdoor playtime?	☐ <10 minutes/day ☐ 10–30 minutes/day ☐ >30 minutes/day
8	Does your child consume vitamin D-rich foods (eggs, fish)?	☐ Yes ☐ No
9	Does your house has windows	☐ Yes ☐ No
10	If yes Q9, Do you open windows for sunlight at home?	☐ Yes ☐ No
SECTION VI: OBSERVATIONAL CHECKLIST		
1	House has open windows	☐ Yes ☐ No
2	Child appears frequently outdoors	☐ Yes ☐ No

ክፍል ፩፡ መለያ

መጠይቅ ኮድ፡ _____

የቃለ መጠይቅ ቀን፡ ____ / ____ / 2026

መኖሪያ ቦታ፡ ☐1 = የከተማ ☐2 = ገጠር

ቀበሌ፡ _____

የቃለ መጠይቅ አድራጊ መታወቂያ ቁጥር፡ _____

ክፍል II፡ ሶሽዮ-ዲሞክራሲ ካራክተርስቲክ

ተ.ቁ	ጥያቄው	መልሶች
1	የእናት ዕድሜ	_____ ዓመት
2	የጋብቻ ሁኔታ	☐ ያገባች ☐ ያላገባች ☐ የፈታች ☐ የሞተባች
3	ሃይማኖት	☐ ኦርቶዶክስ ☐ ሙስሊም ☐ ፕሮቴስታንት ☐ ሌላ _____
4	የትምህርት ሁኔታ	☐ ማንበብና መጻፍ የማትችል ☐ አንደኛ ደረጃ (1-8) ☐ ሁለተኛ ደረጃ (9-12) ☐ ዲፕሎማ+
5	ሥራ	☐ የቤት እመቤት ☐ የመንግስት ሰራተኛ ☐ ነጋዴ ☐ ገበሬ ☐ ሌላ _____
6	የባል የትምህርት ደረጃ	☐ ማንበብና መጻፍ የማይችል ☐ አንደኛ ደረጃ (1-8) ☐ ሁለተኛ ደረጃ (9-12) ☐ ዲፕሎማ+
7	ወርሃዊ የቤተሰብ ገቢ (ETB)	☐ <2000 ☐ 2000-5000 ☐ >5000
8	ከአምስት ዓመት በታች የሆኑ ልጆች ብዛት	_____
9	የትንሹ ልጅ ዕድሜ (ወር)	_____ ወር
10	የልጁ ጾታ	☐ ወንድ ☐ ሴት
11	የትንሹ ልጅ የትውልድ ቅደም ተከተል	☐ 1ኛ ☐ 2ኛ ☐ >3ኛ
12	በባለፈው እርግዝና ወቅት የቅድመ ወሊድ ክትትል ነበረሽ?	☐ አዎ ☐ አይ
13	በባለፈው የወሊድ ግዜ የድህረ ወሊድ ክትትል ነበረሽ?	☐ አዎ ☐ አይ
14	የሚዲያ ተጋላጭነት	☐ አለኝ ☐ የለኝም

ክፍል III፡ ስለ ሽይታሚን ዲ ያለ እውቀት

ተ.ቁ	ጥያቄ	መልስ
1	ሻይታሚን ዲ ለአጥንት ጤና አስፈላጊ ሲሆን ሪኬትስን ይከላከላል።	☐ አዎ ☐ አይ ☐ አላውቅም
2	የፀሐይ ብርሃን ዋናው የሻይታሚን ዲ ምንጭ ነው።	☐ አዎ ☐ አይ ☐ አላውቅም
3	የቀትር ፀሐይ ለደህንነት መጋለጥ አደገኛ ነው።	☐ አዎ ☐ አይ ☐ አላውቅም
4	ጥቁር ቆዳ ለሻይታሚን ዲ ከቀላል ቆዳ ይልቅ ለፀሐይ መጋለጥ ያስፈልገዋል።	☐ አዎ ☐ አይ ☐ አላውቅም
5	የሕፃን ሰውነት በሙሉ መሸፈን የሻይታሚን ዲ እጥረትን ይከላከላል።	☐ አዎ ☐ አይ ☐ አላውቅም
6	በቀን ከ10-30 ደቂቃ የፀሐይ ብርሃን ለህፃናት በቂ ነው።	☐ አዎ ☐ አይ ☐ አላውቅም
7	የፀሐይ ብርሃን ሳይጋለጥ ምግብ ብቻውን በቂ ነው።	☐ አዎ ☐ አይ ☐ አላውቅም
8	ደመናማ የአየር ሁኔታ ሁሉንም የሻይታሚን ዲ ከፀሐይ እንዳይመነጭ ይከላከላል።	☐ አዎ ☐ አይ ☐ አላውቅም
9	የሻይታሚን ዲ እጥረት የአጥንት ድክመትን ብቻ ያስከትላል።	☐ አዎ ☐ አይ ☐ አላውቅም
10	ከስድስት ወር በታች የሆነ ህፃን በቀጥታ ለፀሐይ መጋለጥ አያስፈልገውም።	☐ አዎ ☐ አይ ☐ አላውቅም
11	የአየር ብክለት የቆዳ ሻይታሚን ዲ ውህደትን በእጅጉ ይቀንሳል	☐ አዎ ☐ አይ ☐ አላውቅም

ክፍል IV: ለሻይታሚን ዲ ያለው አመለካከት

1= በጣም አልስማማም፣ 2= አልስማማም፣ 3= ገለልተኛ፣ 4= እስማማለሁ፣ 5= በጥብቅ እስማማለሁ

ተ.ቁ	ጥያቄ	1	2	3	4	5
1	የጠዋት ፀሐይ ለህፃናት ጤና እንደሚጠቅም አምናለሁ።	☐	☐	☐	☐	☐
2	የፀሐይ ብርሃን ለህፃናት የቆዳ ካንሰርን ከማጣት ይልቅ የመጋለጥ እድልን ይጨምራል።	☐	☐	☐	☐	☐
3	ለልጄ ሻይታሚን ዲ በየቀኑ ከቤት ውጭ ጊዜ ማሳለፍ ጠቃሚ ነው።	☐	☐	☐	☐	☐
4	የቀን ፀሐይ ለልጆች ፈጣን የሻይታሚን ዲ ለማግኘት በጣም ጥሩ ነው	☐	☐	☐	☐	☐
5	በቂ የፀሐይ ብርሃን ሲኖር የሻይታሚን ዲ ተጨማሪዎች አያስፈልጉም።	☐	☐	☐	☐	☐
6	ልጆችን ከፀሐይ ብርሃን መጠበቅ ከሪኬትስ አደጋ የተሻለ ነው።	☐	☐	☐	☐	☐
7	የቤተሰቤ ባህል ለጨቅላ ሕፃናት የፀሐይ መጋለጥን ይደግፋል።	☐	☐	☐	☐	☐

8	የማህበረሰብ የጤና ትምህርት በሻይታሚን ላይ ያስፈልጋል	☐	☐	☐	☐	☐
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ክፍል V: ከሻይታሚን ዲ ጋር የተያያዘ ልምዶች

ተ.ቁ	ጥያቄ	መልስ
1	ልጅዎን ለፀሐይ ብርሃን ያጋልጣሉ?	☐ አዎ ☐ አይ
2	አዎ ከሆነ፣ በሳምንት ስንት ቀናት?	☐ በየቀኑ ☐ 3–4 ቀናት ☐ 1–2 ቀናት ☐ አልፎ አልፎ
3	ልጅዎን ለፀሐይ ብርሃን የሚያጋልጡበት ሰዐት	☐ ጠዋት (8–10 ጥዋት) ☐ እኩለ ቀን ☐ ምሽት
4	ልጅዎን ለምን ያህል ደቂቃ ለፀሐይ ያጋልጣሉ?	☐ <10 ደቂቃ ☐ 10–30 ደቂቃ ☐ >30 ደቂቃ
5	ህፃኑ በሚጋለጥበት ጊዜ መሸፈኛ ልብስ ያደርጋል?	☐ አዎ ☐ አይ
6	ጠዋት ላይ ለፀሐይ ሲጋለጡ የፀሐይ መከላከያ ወይም ሌሎች የቆዳ ቅባቶች ይጠቀማሉ?	☐ አዎ ☐ አይ
7	የልጆች ሳምንታዊ የውጪ ጨዋታ ጊዜ?	☐ <10 ደቂቃ/ቀን ☐ 10–30 ደቂቃ/ቀን ☐ >30 ደቂቃ/ቀን
8	ልጅዎ በሻይታሚን ዲ የበለፀጉ ምግቦችን (እንቁላል፣ አሳ) ይመገባል?	☐ አዎ ☐ አይ
9	ቤታቼህ መስኮቶች አሉት?	☐ አዎ ☐ አይ
10	መልስዎ አዎ ከሆነ፣ በቤት ውስጥ የፀሐይ ብርሃን እንዲኖር መስኮቶችን ይከፍታሉ?	☐ አዎ ☐ አይ

ክፍል VI: የምልከታ ዝርዝር

ተ.ቁ	መመልከቻ	መልስ
1	ቤቱ ክፍት መስኮቶች አሉት	☐ አዎ ☐ አይ
2	ህፃኑ ብዙ ጊዜ ከቤት ውጭ ይታያል	☐ አዎ ☐ አይ

አመሰግናለሁ!